

System Analysis & Design: Students Focus Monitoring System using Gaze Estimation

This document outlines the system analysis and design for the "Students Focus Monitoring System using Gaze Estimation" project. The system aims to develop an intelligent computer vision-based solution to track and analyze student attention during online sessions using gaze estimation techniques.

1. Problem Statement & Objectives

Problem Statement

With the rise of online education, maintaining student engagement and monitoring focus during virtual lectures and exams has become a significant challenge. Traditional methods lack real-time, non-invasive mechanisms to track whether a student is actively paying attention or distracted. There is a need for an automated system that can reliably assess student focus without requiring specialized hardware beyond a standard webcam.

Project Goals

The primary goal of this project is to build a scalable and non-invasive monitoring system that leverages Computer Vision, Gaze Estimation, and Machine Learning to determine student focus levels. The system aims to provide real-time monitoring, generate alerts for extended periods of distraction, produce analytical reports for educators, and detect suspicious behavior during online assessments.

Use Case Diagram & Descriptions

While a formal diagram is not provided, the key interactions involve the following actors and use cases:

- **Student:** Interacts with the system via a standard webcam during online sessions. The system captures video streams to extract eye positions and gaze direction.
- **Teacher/Educator:** Receives real-time alerts when students lose attention, accesses analytical reports to gauge lecture engagement, and monitors for suspicious behavior during exams.
- **System Administrator:** Manages the deployment, monitoring (latency, accuracy, data drift), and retraining of the machine learning models.

Functional & Non-Functional Requirements

Functional Requirements:

- The system must capture video input from standard webcams.

- The system must perform face detection and alignment (using MTCNN, RetinaFace, or MediaPipe).
- The system must extract Region of Interest (ROI) including eyes, mouth, and full face.
- The system must calculate visual features such as Eye Aspect Ratio (EAR) for blink detection, PERCLOS, pupil diameter variations, and gaze direction.
- The system must classify the student's state as "Focused" or "Distracted" in real-time.
- The system must generate alerts for teachers when a student loses focus.
- The system must log data and generate analytical reports and interactive dashboards.

Non-Functional Requirements:

- **Performance:** The system must operate in real-time with low latency, optimizing models using ONNX and quantization for lightweight deployment.
- **Scalability:** The system should support deployment as a desktop app, web app, or cloud service.
- **Robustness:** The system must handle high challenge conditions such as lighting variations, different angles, and occlusions (e.g., glasses, head movement).
- **Privacy & Ethics:** The system must adhere to ethical considerations regarding privacy concerns, data protection, and user consent.

Software Architecture

The system follows a pipeline architecture typical of machine learning applications, transitioning from data ingestion to real-time inference:

- 1 **Data Ingestion:** Webcam input captures real-time video.
- 2 **Preprocessing Pipeline:** Frames are extracted, faces are detected and aligned, and lighting variations are normalized.
- 3 **Feature Extraction:** Visual features (eye-based, facial, head pose) and advanced embeddings (Face-Net, VGG-Face) are extracted.
- 4 **Model Inference:** A trained machine learning model (Traditional ML, Deep Learning, or Hybrid) classifies the attention level.
- 5 **Presentation Layer:** A dashboard displays live video, attention level indicators, graphs, and alerts.

2. Database Design & Data Modeling

ER Diagram (Entity-Relationship Diagram) & Schema Considerations

Although a specific database schema is not detailed in the project overview, the system requires structured data storage for logging and reporting. The logical schema would likely include:

- **User/Student Table:** Stores student profiles and consent records.
- **Session Table:** Logs details of the online lecture or exam (Session ID, duration, time of day, difficulty level).

- **Attention Log Table:** Stores time-series data of focus levels, gaze directions, EAR, MAR, and head pose metrics per student per session.
- **Alert Table:** Records instances of extended distraction or suspicious behavior, linked to the specific session and student.

The physical schema must be optimized for fast write operations (time-series data logging) and complex read queries for generating analytical reports and dashboards.

3. Data Flow & System Behavior

Data Flow Diagram (DFD) Context

- 6 **Input:** Video frames flow from the student's webcam into the preprocessing module.
- 7 **Processing:** The preprocessing module extracts frames, detects faces, and normalizes the image. This data flows into the feature extraction module.
- 8 **Feature Engineering:** Extracted features (EAR, gaze direction, head pose) are passed to the machine learning model.
- 9 **Inference:** The model outputs a classification (Focused/Distracted) or an attention score.
- 10 **Output:** The results flow to the dashboard for visualization and to the alert system to notify educators. Data is also logged for future retraining and analysis.

System Behavior (Activity & State Representation)

- **Activity Flow:** The system continuously loops through capturing a frame, preprocessing it, extracting features, running inference, and updating the dashboard. If the attention score drops below a threshold for a specified duration, an alert activity is triggered.
- **State Transitions:** A student's state transitions between "Focused" and "Distracted" based on the model's real-time classification. The system monitors these transitions to calculate metrics like attention decay over time and total distraction time.